

HARSH SAW

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SUMMARY

AI/ML engineer with 2+ years of internship experience shipping production RAG systems, LLM pipelines, and agentic AI workflows. Currently architecting semantic search and conversational AI at OmMuse using AWS Bedrock and MongoDB Vector Store; prior work spans vector embedding pipelines, ML-driven demand forecasting, and distributed backend services serving tens of thousands of users.

EDUCATION

Bachelor of Computer Information Systems
Okanagan College

Expected Dec 2026
Kelowna, BC, Canada

EXPERIENCE

- OmMuse** Seattle, US (Remote)
Software Engineer Intern *Nov 2025 – Present*
- Architecting AI-powered semantic search and natural language chat using a **RAG pipeline** (MongoDB Vector Store, **AWS Bedrock**), enabling producers to query 12K+ user platform with 18K+ tracks and 67 GB of audio conversationally.
 - Shipped features across a polyglot monorepo (**Go gRPC/protobuf, TypeScript/React, Python**) unified by Bazel, owning full AWS deployment — S3, SQS, ECR, Docker, Pulumi IaC, Envoy proxy, and GitHub Actions CI/CD.
 - Engineered a cross-platform desktop uploader (Electron) with folder-sync (chokidar + SQLite WAL), parallelized S3 multipart uploads, OAuth 2.0 PKCE, and typed IPC bridge exposing 70+ gRPC methods to the renderer.
 - Built HubSpot CRM sync in **Go** — async batch worker (sync.Map dedup, 30s flush, 100-user chunks) syncing live metrics across 8 API trigger points.
- MoreThinks Solutions Ltd.** Burnaby, BC
Full-Stack Developer Intern *Jun 2025 – Sept 2025*
- Engineered automated token generation and **vector embedding pipelines**, producing structured data consumed by ML recommendation services; delivered full-stack features via AWS EKS and API Gateway CI/CD.
- Bwisher Ltd.** Remote
Software Engineer Intern *Dec 2024 – Jun 2025*
- Built **FastAPI** services for customer behavior aggregation and **demand forecasting**, generating **ML-driven segmented targeting data** consumed by the Kafka pipeline to optimize offer personalization across 30K+ users.
 - Architected a multi-channel platform using **Apache Kafka + BullMQ**, orchestrating SMS/email/WhatsApp campaigns across **30K+ users** while feeding real-time CRM and analytics pipelines.
 - Designed idempotent NestJS + FastAPI microservices for payment/credit workflows with retry/exponential backoff; owned Jenkins, Kubernetes, Prometheus, and Grafana deployment stack.
- Jythu Ltd.** Remote
Full-Stack Developer Intern *Jan 2024 – Dec 2024*
- Built FFmpeg HLS streaming pipeline (S3/CloudFront + **SQS/Lambda/SNS**) cutting seek time **30s → 1.5s**, bandwidth **–70%**; provisioned full AWS infra with Terraform and GitHub Actions for production apps serving 2K+ learners.

PROJECTS

- Document Intelligence Platform (RAG)** | *FastAPI, LangChain, Qdrant, Groq, Gemini, AWS ECS Fargate* [Demo](#) | [GitHub](#)
- Multi-tenant **RAG platform** with session-isolated **Qdrant** indexes; resilient LLM orchestration via **LangChain RunnableWithFallbacks** (Groq→Gemini failover) with streaming retrieval, context-aware query rewriting, and SHA-256 idempotent ingestion.
 - Deployed to **AWS ECS Fargate** with GitHub Actions CI/CD; hybrid document chunking enabling semantic Q&A over PDF, DOCX, and TXT at scale.
- CodeExpo — Browser-Based IDE & Sandbox** | *React, Node.js, Docker, Nginx, xterm.js, Socket.IO* [Demo](#) | [GitHub](#)
- Full-stack browser IDE with live terminal (xterm.js over WebSocket), real-time code execution, Socket.IO file sync, and Docker multi-stage containerized sandboxes per session.

TECHNICAL SKILLS

Languages: Python, Go (Golang), TypeScript, JavaScript (ES6+), Bash
AI & GenAI: LangChain, LangSmith, AutoGen, RAG, AWS Bedrock, FAISS, Qdrant, Pinecone, AstraDB
Backend & Messaging: FastAPI, NestJS, Express, gRPC/Protobuf, Apache Kafka, BullMQ, GraphQL
Cloud & DevOps: AWS (Bedrock, ECS, S3, SQS, Lambda), Docker, Kubernetes, Terraform, Pulumi, GitHub Actions
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite
Frontend: React, Next.js, React Native, Qwik.js, Electron

ACHIEVEMENTS

- End-to-end stock prediction pipeline: **500 S&P 500 constituents**, 5 yrs of 15-min bar data, XGBoost across 26 intraday horizons on **4x NVIDIA H100 GPUs** (DRAC) with warm-start retraining every 15 min. Top **2%** freelancer on Freelancer.com (5-star, 25+ projects). Won inter-university hackathon.